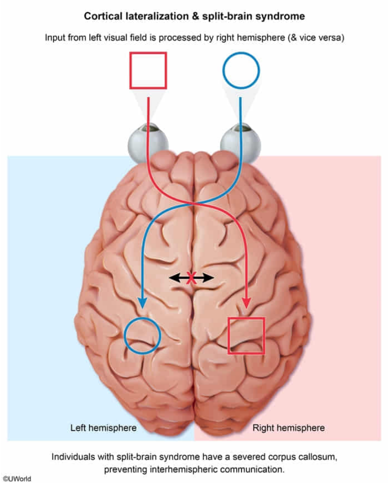
A researcher is testing a patient's neurological function. When a word is flashed briefly in the patient's right visual field, he can correctly vocalize what he saw. When a word is flashed briefly in the patient's left visual field, he is unable to say what he saw but can correctly draw it. Which of the following is most likely damaged in this patient?

- A. Left retina [1%]
- B. Wernicke area [25%]
- C. Right occipital cortex [22%]
- ✔ D. Corpus callosum [50%]

Correct
51% answered correctly

Explanation:



The **corpus callosum** is a bundle of myelinated axonal projections **connecting** the right and left **hemispheres** of the brain, allowing the two hemispheres to communicate. The right and left hemispheres are **specialized** for certain processes, known as **cortical lateralization**. Each hemisphere is responsible for **contralateral control** of the body: The left hemisphere controls touch and movement on the right side of the body, and vice versa. Each hemisphere is also specialized for certain cognitive processes. The **left hemisphere** is specialized for **language** functions, including **speech production** (Broca area) and **language comprehension** (Wernicke area).

Severing the corpus callosum is most frequently used to treat severe epileptic seizures. Individuals with a severed corpus callosum ("split-brain") experience a disconnection between information processed on the right and left sides of the brain. Information presented in a patient's left visual field is processed in the right hemisphere; without interhemispheric communication, the patient is unable to express what is seen using *language* but would be able to draw it.

(Choice A) If the patient's left retina were damaged, he would be unable to receive and transmit information presented to the left visual field. Therefore, he would not be able to process the word or correctly draw it.

(Choice B) The patient is correctly able to vocalize what is seen when it is flashed briefly in the *right* visual field, suggesting that both language centers in the left hemisphere (speech production in Broca area and language comprehension in Wernicke area) are intact.

(Choice C) Damage to the right occipital cortex would result in an inability to process visual information presented in the left visual field. This patient is still able to draw the image represented by the word even though he cannot speak it, suggesting that the occipital lobe is intact.

Educational objective:
The corpus callosum allows communication between the right and left hemispheres of the brain, which are specialized for certain functions (cortical lateralization). Each hemisphere contralaterally controls the opposite side of the body. Language centers are located in the left hemisphere.

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